Volume of prisms

1

a

i Triangle ii $12\,\text{cm}^2$ iii $276\,\text{cm}^3$

b

i Trapezium ii $40\,\text{cm}^2$ iii $240\,\text{cm}^3$

c

i Rectangle ii $24\,\text{m}^2$ iii $288\,\text{m}^3$

d

i Kite ii $16\,\text{cm}^2$ iii $32\,\text{cm}^3$

2

a $216\,\text{cm}^3$ **b** $48\,\text{cm}^3$ **c** $32\,\text{cm}^3$ **d** $217.5\,\text{cm}^3$

e $330\,\text{cm}^3$ **f** $84\,\text{cm}^3$ **g** $270\,\text{cm}^3$

3 $200\,\text{cm}^3$

4 $9600\,\text{mm}^3$

Volume of cylinders

5 $V = \pi r^2 h$

6

a Circle **b** $36\pi\,\text{cm}^2$ **c** $360\pi\,\text{cm}^3$

7

a $10\,995.6\,\text{units}^3$ **b** $2\,413\,754.1\,\text{units}^3$ **c** $785.4\,\text{cm}^3$ **d** $240.3\,\text{cm}^3$

e $45.9\,\text{cm}^3$ **f** $70.7\,\text{cm}^3$ **g** $307.9\,\text{cm}^3$ **h** $1608.5\,\text{units}^3$

i $883.6\,\text{cm}^3$ **j** $263.9\,\text{cm}^3$ **k** $609.5\,\text{cm}^3$

Volume of compound solids

8

a

i A semicircle and a square ii $22.283\,\text{m}^2$

iii $222.83\,\text{m}^3$

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

b

i Two rectangles

ii $34\ \text{cm}^2$

iii $68\ \text{cm}^3$

c

i Two rectangles

ii $40\ \text{cm}^2$

iii $280\ \text{cm}^3$

d

i A triangle and a rectangle

ii $37.5\ \text{m}^2$

iii $412.5\ \text{m}^3$

9

a $399\ \text{cm}^3$ **b** $1080\ \text{cm}^3$ **c** $828\ \text{cm}^3$ **d** $982.35\ \text{m}^3$

e $300\ \text{cm}^3$ **f** $293.52\ \text{cm}^3$

Unknown side given the volume

10 $a = 4\ \text{mm}$

11 $p = 6\ \text{mm}$

12 $k = 11$

13 $y = 10.1$

14 $70\ \text{cm}^2$

15 $9\ \text{cm}$

16 $3\ \text{cm}$

17 $9\ \text{cm}$

Volume of spheres

18

a $113.10\ \text{cm}^3$ **b** $4577.20\ \text{cm}^3$ **c** $113\,550.33\ \text{mm}^3$ **d** $33.51\ \text{cm}^3$

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

19 The student's mistake was substituting the diameter for the radius. Correct volume is $381.70\ \text{cm}^3$

20

a $817.283\ \text{mm}^3$ **b** $2572.441\ \text{cm}^3$ **c** $3.054\ \text{m}^2$ **d** $0.351\ \text{cm}^2$

Applications

21 $2\ \text{m}^3$

22

a $29.69\ \text{cm}^3$ **b** $169.2\ \text{g}$

23 $0.57\ \text{m}^3$

24

a $1335.18\ \text{cm}^3$ **b** $4005.54\ \text{cm}^3$ **c** $28\ \text{L}$

25

a $1198.5\ \text{m}^3$ **b** $299.625\ \text{m}^3$ **c** $h = 0.5875\ \text{m}$

26

a $8151\ \text{cm}^3$ **b** 5

27

a $2\ \text{cm} \times 5\ \text{cm} \times 2\ \text{cm}$ **b** $20\ \text{cm}^3$

c $15\,120\ \text{cm}^3$ **d** 756

28 $d = 29\ \text{cm}$

29 $h = 160\ \text{cm}$

30 $5424.605\ \text{cm}^3$

31 8000 balls

32 $113\,097\ \text{cm}^3$

33 $22\,530.3$ times bigger

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

34 $1.646 \times 10^{11}\ \text{km}^3$

35

a $32.5542\ \text{cm}$ **b** $18\,064\ \text{cm}^3$

36

a $804.2\ \text{cm}^3$ **b** $1206.4\ \text{cm}^3$ **c** 33%

© ClipSAT: By Mr. Mohamed Abdallah +966597688647